GEN 3.5 METEOROLOGICAL SERVICES

1. 책임업무

민간항공에 대한 기상지원 책임기관은 기상청 소속 항공기상청이다.

항공기상청

(우) 22382 인천광역시 영종구 제2터미널대로 444,
제2합동청사

전화 : 032-740-2800
팩스 : 032-740-2817
항공고정통신망 : RKSIYPYX
Website : http://amo.kma.go.kr

다음 문서에 기술된 ICAO 표준 및 권고의 절차와 방법을 적용하고 있다.

부속서 3 : 국제항공항행을 위한 기상업무

국내 규정과 ICAO 국제표준 및 권고실무 사이에 현저한 차이점은 없다.

1. Responsible Service

The authority responsible for the meteorological services for civil aviation are the Aviation Meteorological Office(AMO) of the Korea Meteorological Administration (KMA).

The Aviation Meteorological Office

444, Je2terminal-daero, Yeongjong-gu, Incheon, 22382
Republic of Korea

TEL : +82-32-740-2800
FAX : +82-32-740-2817
AFS : RKSIYPYX
Website : http://amo.kma.go.kr

The service is provided in accordance with the provisions contained in the following ICAO Standards, Recommended Practices and Procedures.

Annex 3 : Meteorological Service for International Air Navigation

There is no significant difference between the National Regulations and the corresponding International Standard, Recommended Practices and Procedures.

2. 관할구역

항공기상청에서는 인천비행정보구역(Incheon FIR)에 관련되는 중요기상정보 및 기타 기상정보 발표의 책임이 있다.

2. Area of responsibility

The Aviation Meteorological Office is responsible for issuing meteorological services and other meteorological information relating to the Incheon FIR.

3. 기상관측 및 보고

3. Meteorological observations and reports

Name of Station/ location Indicator	Frequency & Type of Observation/ automatic observing equipment	Types of MET Report & Availability of trend forecasts	Observation System & Site(s)	Hours of operation (UTC)	Climatological information
1	2	3	4	5	6
Aviation Meteorological Office RKSI (Incheon INTL)	Half hourly routine observation automatic : AVBL	METAR TREND	a) AMOS ・Transmissometer : Both ends & MID per each RWY ・Ceilometer : Both ends of RWY 15L/33R, 16L/34R ・Ultrasonic wind sensor : Both ends of each RWY ・Barometer : Both ends of RWY 15L/33R, 16L/34R b) LLWAS : 12 sites around airport c) TDWR : 6.7 km W from ARP	H24	Climatological table AVBL
Gimpo Airport Weather Office RKSS (Seoul/Gimpo INTL)	Hourly routine plus special observation automatic : AVBL	METAR SPECI TREND	a) AMOS ・Transmissometer : Both ends & MID per each RWY ・Ceilometer : Both ends of RWY 14R/32L ・Ultrasonic wind sensor : Both ends of each RWY ・Barometer : RWY 14R, 32R	H24	Climatological table AVBL
Jeju Airport Weather Office RKPC (Jeju INTL)	Hourly routine plus special observation automatic : AVBL	METAR SPECI TREND	a) AMOS ・Transmissometer : Both ends & MID per each RWY 07/25 ・Ceilometer : Both ends of RWY 07/25 ・Ultrasonic wind sensor : Both ends of each RWY ・Barometer : Both ends of RWY 07/25 b) LLWAS : 11 sites around airport c) TDWL : Two TDWLs are operation at airport's vicinity, located 0.5 km NW, 2.4 km SW from ARP	H24	Climatological table AVBL

Change : Information of address, TEL of authority and Amended phrase(telefax → FAX).

Name of Station/ location Indicator	Frequency & Type of Observation/ automatic observing equipment	Types of MET Report & Availability of trend forecasts	Observation System & Site(s)	Hours of operation (UTC)	Climatological information
1	2	3	4	5	6
Gimhae Airforce Weather Office RKPK (Busan/Gimhae INTL)	Hourly routine plus special observation automatic : AVBL	METAR SPECI	a) AMOS · Transmissometer : Both ends of RWY & MID of RWY 18R/36L · Ceilometer : Both ends of RWY 18R/36L · Barometer: Ends of RWY 36L	H24	NIL
Cheongju Airforce MET Office RKTU (Cheongju INTL)	Hourly routine plus special observation automatic : AVBL	METAR SPECI	a) AMOS · Transmissometer : Both ends of RWY 24R/06L · Ceilometer : Both ends of RWY 24R/06L · Barometer : Ends of RWY 24R	H24	NIL
Yangyang Airport Weather Station RKNY (Yangyang INTL)	Hourly routine plus special observation automatic : AVBL	METAR SPECI TREND: Aerodrome operating hours	a) AMOS · Transmissometer : Both ends of RWY · Ceilometer(Laser type) : Both ends of RWY · Ultrasonic wind sensor : Both ends of RWY · Barometer : Both ends of RWY b) LLWAS : 8 sites around airport	Aerodrome operating hours	Climatological table AVBL
Daegu Airforce MET Office RKTN (Daegu INTL)	Hourly routine plus special observation automatic : AVBL	METAR SPECI	a) AMOS · Transmissometer : Both ends of RWY 31L/13R · Ceilometer : Both ends of RWY 31L/13R · Barometer : Ends of RWY 31L	H24	NIL
Muan Airport Weather Office RKJB (Muan INTL)	Hourly routine plus special observation automatic : AVBL	METAR SPECI TREND: Aerodrome operating hours	a) AMOS · Transmissometer : Both ends of RWY · Ceilometer(Laser type) : Both ends of RWY · Ultrasonic wind sensor : Both ends of RWY · Barometer : Both ends of RWY	H24	Climatological table AVBL
Gwangju Airforce MET Office RKJJ	Hourly routine plus special observation automatic : AVBL	METAR SPECI	a) AMOS · Transmissometer : Both ends of RWY · Ceilometer : Both ends of RWY 04L/22R · Barometer : Ends of RWY 04L	H24	NIL
Yeosu Airport Weather Station RKJY (Yeosu domestic)	Hourly routine plus special observation automatic : AVBL	METAR SPECI TREND: Aerodrome operating hours	a) AMOS · Transmissometer : Both ends of RWY · Ceilometer(Laser type) : Both ends of RWY · Ultrasonic wind sensor : Both ends of RWY · Barometer : Both ends of RWY	Aerodrome operating hours	Climatological table AVBL
Wonju Airforce MET Office RKNW	Hourly routine plus special observation automatic : AVBL	METAR SPECI	a) AMOS · Transmissometer : ends of RWY · Ceilometer : Both ends of RWY 03/21 · Barometer : Ends of RWY 03	H24	NIL
Sacheon Airforce MET Office RKPS	Hourly routine plus special observation automatic : AVBL	METAR SPECI	a) AMOS · Transmissometer : Both ends of RWY · Ceilometer : Both ends of RWY 06R/24L · Barometer : Ends of RWY 06R	H24	NIL
Ulsan Airport Weather Office RKPU (Ulsan domestic)	Hourly routine plus special observation automatic : AVBL	METAR SPECI TREND: Aerodrome operating hours	a) AMOS · Transmissometer : Both ends of RWY · Ceilometer : Both ends of RWY · Ultrasonic wind sensor : Both ends of RWY · Barometer : Both ends of RWY	H24	Climatological table AVBL
Pohang-Gyeongju Navy MET Office RKTH	Hourly routine plus special observation automatic : AVBL	METAR SPECI	a) AMOS · Transmissometer : Both ends of RWY · Ceilometer : Both ends of RWY · Wind seonsor : Both ends of RWY · Barometer : Both ends of RWY · TDWR(WRK-200)	H24	NIL

Remarks

· AMOS : Aerodrome Meteorological Observation System
공항기상관측장비
- Transmissometer, Ceilometer (Laser type), Ultrasonic wind sensor, Barometer
시정계(RVR), 운고계, 초음파풍향·풍속계, 기압계
· LLWAS : Low-Level Wind shear Alert System
저층급변풍경고장비
· TDWR : Terminal Doppler Weather Radar
공항기상레이더
· TDWL : Terminal Doppler Wind Lidar
공항기상라이다
* At the time of reports, the visual observations are added to AMOS data manually.

Change : Information of MET observations and reports for RKPK, RKTU, RKTN, RKJJ, RKPS and Establishment of MET office for RKNW.

4. 지원업무 내용

국제선을 운항하는 모든 항공기에는 비행예보철이 지원된다. 비행예보철에는 세계공역예보시스템으로부터 수신하거나 항공기상청이 생산한 공항예보, 중요기상예보, 상층바람, 기온예상도 및 SIGMET/AIRMET 정보가 포함된다.

국내선 항공기와 헬리콥터에 대해서는 국내 및 국제 공항과 관심 지역에 대한 최근 공항관측, 공항예보, 공항경보 및 급변풍경보, 저고도 공역예보, SIGMET/AIRMET 정보가 제공된다.

기상정보는 항공고정통신망, FAX 및 인터넷을 통하여 제공된다. 비행예보철과 부가적 또는 특정한 정보를 인터넷으로 제공받고자할 경우에는 항공기상청과 이용자 간에 협약이 있어야 한다.

기상에 관한 브리핑과 자문은 각 공항의 기상관서에서 제공된다. 그러나 여건상 불가능한 경우에는 항공기상청이 전화 또는 전자우편을 통한 브리핑과 자문을 행할 수 있다.

공역예보는 항공기상청에서 발표하며 국내 비행 시 사용을 위해 항공기상청 홈페이지를 통하여 게시된다.

5. 항공사가 고지하여야 할 사항

비행예보철을 필요로 하는 운항 관계자는 다음의 경우 1일 전에 항공기상청에 통보해야 한다.

a. 신규노선 또는 새로운 종류의 운항 계획이 있을 때
b. 기존 정기운항내용의 변경 사항이 있을 때
c. 기상업무 제공에 영향을 주는 기타 변경사항이 있을 때

항공사는 다음 사항이 발생했을 때 적어도 출발 6시간 전에 항공기상청에 통보해야한다.

a. 부정기 항공기가 운항할 때
b. 항공기 운항이 지연, 조기 출발 또는 취소될 때
c. 비행예정표

6. 항공기 관측보고

공대지 데이터 링크를 사용하거나, 자동종속감시시설(ADS-C) 또는 2차 감시레이더(SSR)의 Mode S를 적용할 때, 비행 중에는 매 15분마다 그리고 비행 후 최초 10분 동안의 상승 단계에서는 매 30초마다 자동 정기 관측을 해야 한다. 그러나 음성통신으로 보고가 이루어지거나 공대지 데이터 링크, ADS-C 또는 SSR Mode S를 통한 보고를 적용할 수 없을 때, 정기항공기관측은 아시아·태평양 구역 내의 기상보고지점에서 보고된다.

4. Types of services

Flight documentation is provided to all aircraft for international flights. The documentation normally contains aerodrome forecasts(TAF code form), significant weather forecasts, upper wind and temperature charts and SIGMET/AIRMET information received from the World Area Forecast System or produced by the AMO.

For domestic flights and helicopter operations, the latest aerodrome reports (METAR, SPECI), and aerodrome forecasts, aerodrome warnings and wind shear warnings, area forecasts for low-level flights and SIGMET/AIRMET information are provided for the domestic and international airports, as well as nation-wide area concerned.

The meteorological information is provided via AFTN, fax and the Internet. When the internet service of flight documentation and other additional or specific information are needed, there must be an agreement between the AMO and users.

Briefing and consultation of meteorological information are, in general, provided by each meteorological office/station at the airport. However, when the local circumstances make it impracticable, the briefing and consultation shall be carried out by the AMO through telephone or e-mail.

Area forecasts are issued by the AMO and disseminated through the AMO website for only domestic use.

5. Notification Required from Operators

The operator or a flight crew member requiring service (flight documentation) shall notify the AMO one day in advance, when :

1. new routes or new types of operation are planned;
2. changes are to be made in the scheduled operations;
3. other changes, affecting the provision of meteorological service, are planned.

The operator or a flight crew member shall notify the AMO at least 6 hours before the expected time of departure, when :

1. non-scheduled flights are to be operated;
2. flights are delayed, advanced or cancelled.
3. flight schedules

6. Aircraft Reports

When air-ground data link is used and automatic dependent surveillance - contract (ADS-C) or secondary surveillance radar (SSR) Mode S is being applied, automated routine observations should be made every 15 minutes during the en-route phase and every 30 seconds during the climb-out phase for the first 10 minutes of the flight. However, when voice communications are used and air-ground data link. ADS-C, SSR Mode reports are not available, routine aircraft observations (AIREPs) are required at the meteorological reporting points in the Asia/Pacific region.

Change : Page control.

7. VOLMET 제공업무

※ 항공기상청 해당사항 없음

7. VOLMET Service

N/A to the AMO.

8. SIGMET과 AIRMET 정보 제공업무

8. SIGMET and AIRMET Service

Name of MWO/location indication	Hours	FIR or CTA served	Type of SIGMET/ validity	Specific SIGMET procedures	AIRMET procedures	ATS unit served	Additional information
1	2	3	4	5	6	7	8
INCHEON/ RKSI	H24	Incheon FIR	SIGMET/ 4 HR	SIGMET VA/TC: VALIDITY 6 HR	AIRMET VALIDITY 4 HR	Incheon ACC, Daegu ACC	NIL

8.1 일반사항

항공교통의 안전을 위한 기상감시소로써 항공기상청은 인천비행정보구역 내에서 비행에 영향을 끼칠 수 있는 기상현상에 대한 지속적인 감시를 수행하고, 필요한 경우 SIGMET정보, AIRMET정보, 공항경보, 급변풍경보를 발표한다.

8.1 General

For the safety of air traffic, the Aviation Meteorological Office(AMO), as the Meteorological Watch Office(MWO), maintains a continuous watch over meteorological conditions affecting flight operations within the Incheon FIR and when necessary, issues the SIGMET and AIRMET information, aerodrome warnings and wind shear warnings.

8.2 기상감시

항공기상청은 다음과 같은 기상감시를 수행한다.

a) 책임구역 내에서 비행에 영향을 미치는 기상상태에 대한 감시

b) 책임구역과 관련되는 위험기상정보 및 기타 정보의 작성

c) 요청이 있을시 위험기상 정보 및 기타 기상정보를 관련 항공교통업무기관에 제공

d) SIGMET 정보와 AIRMET 정보 전파

e) SIGMET 정보로 아직 발표되지 않은 분출전 화산활동, 화산분출 및 화산재 구름에 관해 입수한 정보를 기상당국과 항공교통업무당국 간 합의에 따라 비행정보센터 또는 지역관제센터에 그리고 지역항공항행협정에 의해 결정된 화산재주의보센터에 제공

f) 감시 또는 인근 구역 내 대기로 방사성 물질 방출사고에 관하여 수신한 정보를 기상당국과 항공교통업무당국 간 합의에 따라 비행정보기관 또는 항공교통업무기관에 제공

항공기상청은 ICAO Annex 3, 7장에 기초하여 SIGMET과 AIRMET정보를 발표한다.

8.2 Meteorological Watch

The meteorological watch is performed by the AMO, as follows:

a) maintain watch over meteorological conditions affecting flight operations within its area of responsibility

b) prepare SIGMET and other information relating to its area of responsibility

c) supply SIGMET information and, as required, other meteorological information to associated air traffic services units

d) disseminate SIGMET and AIRMET information

e) supply information received on pre-eruption volcanic activity, a volcanic eruption and volcanic ash cloud for which a SIGMET has not already been issued, to its associated ACC/FIC, as agreed between the meteorological and ATS authorities concerned, and to its associated VAAC as determined by regional air navigation agreement and

f) supply information received concerning the accidental release of radioactive materials into the atmosphere, in the area for which it maintains watch or adjacent areas, to aeronautical information service units and ATS authorities, as agreed between the meteorological and ATS authorities concerned.

The AMO issues SIGMET and AIRMET information in accordance with ICAO Annex 3, Chapter 7.

Change : Page control.

8.3 공항경보와 급변풍경보

항공기상청 또는 각 공항기상관서는 아래에 열거한 현상 중 한 가지 이상이 발생할 것으로 예상될 경우, 계류 중인 항공기와 공항시설들을 보호하기 위해 공항경보를 발표한다.

- 태풍
- 뇌우
- 우박
- 대설
- 강풍
- 호우
- 구름고도(Ceiling : 하늘의 5/8 이상을 가리는 최하층 구름고도)
- 저시정
- 먼지 또는 모래보라
- 어는 강수
- 서리
- 화산재

활주로와 활주로 고도 위 500 m(1 600 ft) 사이의 접근/이륙로 또는 선회접근 중인 항공기 그리고 착륙 또는 이륙을 위해 주행 중인 항공기에 악영향을 미칠 수 있는 급변풍의 관측 또는 예상될 때, 급변풍경보를 발표한다.

공항경보와 급변풍경보는 영어로 발표하며, 지역적으로 합의된 분배목록에 따라서 전파된다.

8.4 비행중인 항공기의 SIGMET 분배

※ 항공기상청 해당사항 없음

8.3 Aerodrome Warnings and Wind Shear Warnings

Aerodrome warnings for the protection of parked aircraft or of other equipment at the airport are issued by the AMO or each airport meteorological office concerned, if one or several of the following phenomena are expected to occur at the airport.

- Typhoon
- Thunderstorm
- Hail
- Heavy snow
- Strong surface wind and gusts
- Heavy rain
- Ceiling
- Low visibility
- Rising sand or dust
- Freezing precipitation
- Frost
- Volcanic ash

Wind shear warnings are issued when wind shear which could adversely affect aircraft on the approach path or take off path or during circling approach between runway level and 500 m(1 600 ft) above that level is observed or expected.
The aerodrome warnings and wind shear warnings are issued in English and are distributed in accordance with a distribution list agreed upon locally.

8.4 Dissemination of SIGMET Information to Aircraft in Flight

※ Not authorized to the AMO.

9. 인터넷 서비스 9. Internet Services

정보명 Service name	유효정보 Information available	서비스 구역, 항공로 및 공항의 범위 Area, Route and Aerodrome covered	전화, 팩스번호 Telephone, FAX numbers Remarks
1	2	3	4
항공날씨 Weather Information Service for Air Navigation Support	기상관측 및 보고 정시 및 특별 관측전문(경향예보 포함) 정시 및 특별 국지관측전문 Meteorological Observations and Reports METAR/SPECI (Including Trend forecast) MET REPORT, SPECIAL Report	인천비행정보구역 (공항예보 : 항공고정통신망과 세계공역예보시스템으로부터 수신된 모든 구역 포함) Incheon FIR (TAF : Including all of the received regions from AFTN and WAFS)	항공기상청 The Aviation Meteorological Office TEL : +82-32-222-3030 FAX : +82-32-740-2817 https://global.amo.go.kr
	항공기 특별 관측보고 Special AIREP		
	항공예보 공항예보, 이륙예보, SIGWX(저고도, 중고도) Aerodrome Forecast TAF, Take-off forecast, SIGWX(Low-level/Medium-level)		
	특보 SIGMET 정보, AIRMET 정보, 공항경보, 급변풍 경보 및 경고, 태풍정보 SIGMET and AIRMET information, Aerodrome warnings, Wind shear warnings and alerts, Typhoon information		
	일기도/WAFS 한반도 WINTEM, 난류예측일기도, 분석일기도, 예상일기도, WAFS 수치예측일기도 Numerical model Weather charts/WAFS charts Domestic WINTEM, Turbulence prediction chart, Synoptic analysis chart, Numerical prediction chart WAFS products (SIGWX, WINTEM, Tropopause, Max Wind, Humidity, Cross Section, Geopotential Height, CAT, CB, ICING, WAFS OPMET)	인천비행정보구역, 세계공역예보시스템으로부터 수신된 모든 구역 Incheon FIR, All of the received regions from WAFS	
	날씨영상 레이더(공항기상레이더, 기상레이더), 낙뢰, 위성영상 Weather images Radar(TDWR, Weather radar), Lightning, Satellite images	인천 비행정보구역 Incheon FIR	
	운항지원 공항기상정보, 항공교통기상정보, 화산재주의보, 우주기상주의보, AMOS 실시간 관측자료 Flight weather information Airport weather information, Meteorological information for air traffic management, Volcanic ash and space weather advisory, AMOS observation data	인천 비행정보구역, 항공고정통신망에서 수신된 모든 구역 Incheon FIR, All of the received regions from AFTN	
항공기상청 홈페이지 Aviation Meteorological Information Service	기상관측 및 보고 정시 및 특별 관측전문(경향예보 포함) Meteorological Observations and Reports METAR/SPECI (Including Trend forecast)	한국, 인천비행정보구역 (공항예보 : 항공고정통신망과 세계공역예보시스템으로부터 수신된 모든 구역 포함) Korea, Incheon FIR (TAF : Including all of the received regions from AFTN and WAFS)	항공기상청 The Aviation Meteorological Office TEL : +82-32-222-3030 FAX : +82-32-740-2817 https://amo.kma.go.kr
	항공예보 공항예보 Aerodrome Forecast TAF		
	공항경보, 급변풍경보 일기도, 위성영상, 레이더영상, 낙뢰정보 Aerodrome warning, Wind shear warning, Weather Chart, Satellite image, Radar image, Lightning information		

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